

Plotting with Scilab is Simple



This article discusses various types of plots based on mathematical computations done in Scilab, which is open source software for numerical computation. Seven different types of plots used in seven different computations are explored, along with illustrations.

Scilab (<http://www.scilab.org>) is open source software for numerical computation with a syntax similar to MathWorks' MATLAB. Scilab has very good plotting capabilities, but some plot types are not available as primitives or functions. So, it's necessary to write some code to obtain them. Note that the keywords of Scilab are categorised as *primitives*, *commands*, *variables*, *functions* and *xcos functions*. These words will be used only in reference to the keywords. To avoid any confusion, the word 'variable' in the sense of a physical quantity (or chemical, graphical ...) will be replaced by the word 'parameter'. In this article, seven new (or almost new) plot types are covered with some practical applications, all made with Scilab 5.5.1 under Linux Mint 17 Xfce. Because the full code for each example is too long for a printed article, it's freely (as in freedom and in beer) available at <https://github.com/astonfe/scilab>

Area plot

In this first example, we need to graphically compare two series of experimental data. These two columns are the per cent recoveries of 13 chemical compounds extracted from environmental samples using two different techniques. We can plot the data with the function *bar* or with an area plot

(or a wide range of other tools, but this is only an example), which requires a bit of coding. This is easy to do because each area can be defined by a combination of the function *plot* and the primitive *xfpoly*, which is a filled polygon with the base at $y=0$ (see the parameter 'base').

Bar plot 3D

This example is about analysing particulate matter in urban air. The parameters are three: *concentration*, *month of the year* and *monitoring station*. Each measure is a point in this three-dimensional space. The function *hist3d* can be an option, but this plot type is not very nice (it's a bit ugly). Even in this case, some coding is required. The plot is built by scanning the data matrix with two nested loops. Each parallelepiped has a square base with the side 'w' equal to 0.8 and the height 'z' equal to the concentration. The parameter *col* is used to scale the colours by aligning the data and the colour map. Note that in Scilab, it is possible to use special characters directly in SciNotes, the built-in editor. SciNotes is a powerful text editor and it also has interesting LaTeX capabilities (some examples will be shown in the following sections about impulse and ternary plots). At present, it's not possible to automatically align the